| | | |
|---|---|---|
| **1(a)** | $h = 0.5$ | B1 |
| | $A \approx \frac{1}{2} \times \frac{1}{2}\{2.287 + 2.834 + 2(4.470 + 6.719 + 7.291)\}$ | M1 |
| | = awrt 10.52 | A1 |
| | | **(3)** |
| **(b)(i)** | $A \approx$ "10.52" – 4 | M1 |
| | = 6.52 | A1ft |
| **(b)(ii)** | $A \approx$ "10.52" | B1ft |
| | | **(3)** |
| | | **Total 6** |

| | | |
|---|---|---|
| **2(a)** | $(S=)\ 6x^2+6xh+2xh$ | B1 |
| | eg $V=3x^2h=972\Rightarrow h=\dfrac{972}{3x^2}$ or eg $hx=\dfrac{324}{x}$<br>$\Rightarrow (S=)\ 6x^2+8x\left(\dfrac{972}{3x^2}\right)$ or $\Rightarrow (S=)\ 6x^2+8\left(\dfrac{324}{x}\right)$ | M1 |
| | $S=6x^2+\dfrac{2592}{x}$* | A1* |
| | | **(3)** |
| **(b)** | $\left(\dfrac{\mathrm{d}S}{\mathrm{d}x}=\right)12x-\dfrac{2592}{x^2}$ | B1 |
| | | **(1)** |
| **(c)** | $12x-\dfrac{2592}{x^2}=0\Rightarrow 12x^3=2592$<br>$\Rightarrow x=\sqrt[3]{\dfrac{2592}{12}}$ | M1 |
| | $x=6$ | A1 |
| | | **(2)** |
| **(d)** | $\left(\dfrac{\mathrm{d}^2S}{\mathrm{d}x^2}=\right)12+\dfrac{5184}{x^3}$ | B1ft |
| | $\dfrac{\mathrm{d}^2S}{\mathrm{d}x^2}>0$ when $x=6$ so minimum | B1 |
| | | **(2)** |
| **(e)** | $S=6(6)^2+\dfrac{2592}{6}=648\ (\mathrm{cm}^2)$ | B1 |
| | | **(1)** |
| | | **Total 9** |

| | | |
|---|---|---|
| **3(a)** | $\left(2+\frac{kx}{8}\right)^7 = 2^7 + \binom{7}{1}2^6\left(\frac{kx}{8}\right) + \binom{7}{2}2^5\left(\frac{kx}{8}\right)^2 + \binom{7}{3}2^4\left(\frac{kx}{8}\right)^3 + ...$ | M1 |
| | $= 128 + 56kx + \frac{21}{2}k^2x^2 + \frac{35}{32}k^3x^3 + ...$ | B1A1A1 |
| | | **(4)** |
| **(b)** | $\frac{35}{32}k^3 - \frac{21}{2}k^2 = \frac{21}{2}k^2 - 56k$ | M1 |
| | $5k^2 - 96k + 256 = 0 \Rightarrow k = ...$ | dM1 |
| | $k = 16, \frac{16}{5}$ | A1 |
| | | **(3)** |
| | | **Total 7** |

| | | |
|---|---|---|
| **4(i)** | E.g. $2=\log_3 9$<br>Eg $\log_3(4x)-\log_3(5x+7)=\log_3\dfrac{4x}{5x+7}$ | M1 |
| | E.g.<br>$36x=5x+7$<br>$\dfrac{4x}{5x+7}=\dfrac{1}{9}$ | A1 |
| | $x=\dfrac{7}{31}$ | A1 |
| | | **(3)** |
| **(ii)** | $\left(\sum_{r=1}^{2}\log_a y^r=\right)\log_a y+\log_a y^2$ or $\left(\sum_{r=1}^{2}(\log_a y)^r=\right)\log_a y+(\log_a y)^2$ | B1 |
| | $\log_a y+\log_a y^2=\log_a y+(\log_a y)^2\Rightarrow 2\log_a y-(\log_a y)^2=0$<br>$\Rightarrow \log_a y(2-\log_a y)=0\Rightarrow \log_a y=2$ | M1 |
| | $y=a^2$ | A1 |
| | | **(3)** |
| | | **Total 6** |

| | | |
|---|---|---|
| **5(a)** | $3^3+3^2(p+3)-3+q=0$ | M1 |
| | eg $\Rightarrow 27+9p+27-3+q=0 \Rightarrow 9p+q=-51$* | A1* |
| | | **(2)** |
| **(b)** | $(-p)^3+(p+3)(-p)^2-(-p)+q=9$ | M1 |
| | eg $-p^3+p^3+3p^2+p+q=9 \Rightarrow 3p^2+p+q-9=0$* | A1* |
| | | **(2)** |
| **(c)** | $3p^2+p+q-9=0$<br>$\Rightarrow 3p^2+p-51-9p-9=0$<br>$\Rightarrow 3p^2-8p-60=0$ | M1 |
| | $p=6$ | A1 |
| | $q=-51-9p=-105$ | A1 |
| | | **(3)** |
| **(d)** | $f(x)=x^3+9x^2-x-105$<br>$f(x)=(x-3)(...x^2+...x+...)$ | M1 |
| | $g(x)=x^2+12x+35$ | A1 |
| | | **(2)** |
| | | **Total 9** |

| | | |
|---|---|---|
| **6(a)(i)** | Centre is (–4, 2) | B1 |
| **(ii)** | $x^2+y^2+8x-4y=0$<br>$\Rightarrow (x+4)^2+(y-2)^2-16-4=0 \Rightarrow r=...$ | M1 |
| | $r=2\sqrt{5}$ oe | A1 |
| | | **(3)** |
| **(b)** | $x^2+y^2+8x-4y=0,\ x+2y+10=0$<br>$\Rightarrow (-2y-10)^2+y^2+8(-2y-10)-4y=0$<br>or<br>$x^2+\left(\frac{-x-10}{2}\right)^2+8x-4\left(\frac{-x-10}{2}\right)=0$ | M1 |
| | $y^2+4y+4=0$ or e.g. $x^2+12x+36=0$ | A1 |
| | $(y+2)^2=0 \Rightarrow y=...$ or e.g. $(x+6)^2=0 \Rightarrow x=...$ | dM1 |
| | (–6, –2) | A1 |
| | | **(4)** |
| **(c)** | $x+2y+10=0 \Rightarrow m_T=-\frac{1}{2} \Rightarrow m_N=2$ or $m_N=\frac{"2"-("-2")}{"-4"-("-6")}$ | M1 |
| | Usually either $y-2=2(x+4)$ or $y+2=2(x+6)$ | dM1 |
| | $y=2x+10$ | A1 |
| | | **(3)** |
| | | **Total 10** |

| | | |
|---|---|---|
| **7(a)** | $r = \sqrt{\frac{12.8}{20}} = 0.8*$<br>or e.g.<br>$20 \times 0.8 \times 0.8 = 12.8$<br>so $r = 0.8*$ | B1* |
| | | **(1)** |
| **(b)** | $a = 20 \div 0.8^2$ | M1 |
| | $= 31.25$ | A1 |
| | | **(2)** |
| **(c)** | $\frac{31.25\left(1-0.8^n\right)}{1-0.8} > 156$ | M1 |
| | Eg $1-0.8^n > 0.9984$<br>$\Rightarrow 0.8^n < 0.0016$ | dM1 |
| | $0.8^n < 0.0016 \Rightarrow n > \frac{\log\left(0.0016\right)}{\log(0.8)}$<br>or<br>$0.8^n < 0.0016 \Rightarrow n > \log_{0.8} 0.0016$ | M1 |
| | $n = 29$ | A1 |
| | | **(4)** |
| | | **Total 7** |

| | | |
|---|---|---|
| **8(i)** | $5\sin(3x+0.1)+2=0$<br>$\Rightarrow 5\sin(3x+0.1)=-2$<br>$\Rightarrow \sin(3x+0.1)=-\frac{2}{5}$ | M1 |
| | $\sin(3x+0.1)=-\frac{2}{5}$<br>$\Rightarrow 3x+0.1=\sin^{-1}\left(-\frac{2}{5}\right)$<br>$\Rightarrow x=\frac{\sin^{-1}\left(-\frac{2}{5}\right)-0.1}{3}$ | dM1 |
| | $x=-0.94,\ -0.17,\ 1.15,\ 1.92$ | A1A1 |
| | | **(4)** |
| **(ii)** | $2\tan\theta\sin\theta=\cos\theta+5$<br>$\Rightarrow 2\sin^2\theta=\cos^2\theta+5\cos\theta$ | M1 |
| | $\Rightarrow 2\left(1-\cos^2\theta\right)=\cos^2\theta+5\cos\theta$ | M1 |
| | $\Rightarrow 3\cos^2\theta+5\cos\theta-2=0$ | A1 |
| | $\cos\theta=\frac{1}{3}\left(,-2\right)$<br>$\Rightarrow \theta=\cos^{-1}\left(\frac{1}{3}\right)=...$ | M1 |
| | $\left(\theta=\right)\ 70.5°,\ 289.5°$ | A1 |
| | | **(5)** |
| | | **Total 9** |

| | | |
|---|---|---|
| **9(a)** | $(0, 5)$ | B1 |
| | | **(1)** |
| **(b)** | $x^2-4x+5=2 \Rightarrow x^2-4x+3=0$<br>$\Rightarrow x=...$ | M1 |
| | $x(E)=1,\ \ x(F)=3$ | A1 |
| | | **(2)** |
| **(c)** | Area $R_1 = \int_0^1 \left(x^2-4x+5-2\right)\mathrm{d}x = \left[\frac{x^3}{3}-2x^2+3x\right]_0^1 = \frac{1}{3}-2+3-0=\frac{4}{3}$ | M1A1 |
| | Area $R_2 = \frac{1}{2}\times 3\times 3 - "\frac{4}{3}" = \frac{19}{6}$<br>or<br>$\int_0^1 \left(5-x-\left(x^2-4x+5\right)\right)\mathrm{d}x+\frac{1}{2}\times 2\times 2 = \left[\frac{3x^2}{2}-\frac{x^3}{3}\right]_0^1+2=\frac{3}{2}-\frac{1}{3}+2=\frac{7}{6}+2=\frac{19}{6}$<br>or<br>$\frac{1}{2}\times 3\times 3-\int_0^1 \left(x^2-4x+5-2\right)\mathrm{d}x=\frac{9}{2}-\left[\frac{x^3}{3}-2x^2+3x\right]_0^1=\frac{9}{2}-\left(\left(\frac{1}{3}-2+3\right)-0\right)=\frac{19}{6}$ | M1A1 |
| | $\frac{\text{area of } R_1}{\text{area of } R_2}=\frac{\frac{4}{3}}{\frac{19}{6}}=\frac{8}{19}$ | A1 |
| | | **(5)** |
| | | **Total 8** |

| | | |
|---|---|---|
| **10** | Considers another relevant case for $n$<br>e.g. $n=3k-1$ or $3k+1$ or $3k+2$<br>For example $n = 3k + 1$<br>$\Rightarrow 2n^2+n+1=2(3k+1)^2+(3k+1)+1=18k^2+15k+4$ | M1 |
| | $=3(6k^2+5k+1)+1$ which is not divisible by 3 | A1 |
| | Considers a third relevant case to complete the 3 cases for $n$ e.g. one of<br>$n=3k-1$ or $3k+1$ or $3k+2$<br>For example $n = 3k - 1$<br>$\Rightarrow 2n^2+n+1=2(3k-1)^2+(3k-1)+1=18k^2-9k+2$ | dM1 |
| | $=3(6k^2-3k)+2$ which is not divisible by 3<br>Hence $2n^2+n+1$ is not divisible by 3 | A1 |
| | | **(4)** |
| | | **Total 4** |